

MEMORY HIERARCHY

Organization of Memory in Computer Systems

Digital Logic and Computer Organization (DLCO) | Unit IV

JNTUK R23 Regulation | Department of ACSE

Presented by (Team of 5 Members):

- Member 1: Karthikeya (Intro & Hierarchy Structure)
- Member 2: Nikhil (Semiconductor RAM & ROM)
- Member 3: Teja (Cache Memory & Performance)
- Member 4: Srinivas (Virtual Memory & Management)
- Member 5: Manohar (Secondary Storage & Summary)



Introduction to Memory Organization

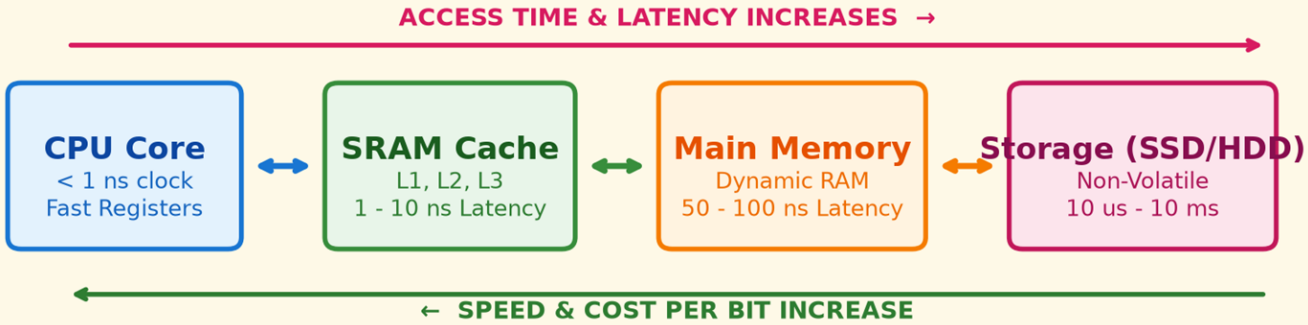
Presented by: Member 1 (Karthikeya) | Basic Concepts & Trade-offs

Role of Memory in Systems

- Stores active instructions & computation data.
- Directly bounds CPU execution throughput.
- **Trade-off:** No single tech gives top speed & capacity.

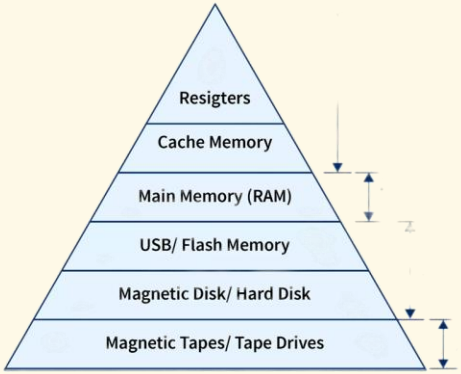
Core Hierarchy Principles

- **Speed vs. Cost:** Faster memory = higher cost per bit.
- **Capacity vs. Speed:** Larger capacity = higher access latency.
- **Tiered Hierarchy:** Blends fast SRAM caches with cheap, mass storage.



Structure of Memory Hierarchy

Presented by: Member 1 (Karthikeya) | The Memory Pyramid



Memory Pyramid Takeaways

- **Registers (< 1 ns):** Internal CPU; fastest, most expensive.
- **Cache SRAM (1-10 ns):** L1-L3 buffers bridging CPU & DRAM.
- **Main DRAM (50-100 ns):** Working system RAM.
- **Secondary Storage (10 us-10 ms):** SSD/HDD mass storage.
- **Trend:** Downward = speed drops, capacity & affordability jump.

Semiconductor RAM Memories

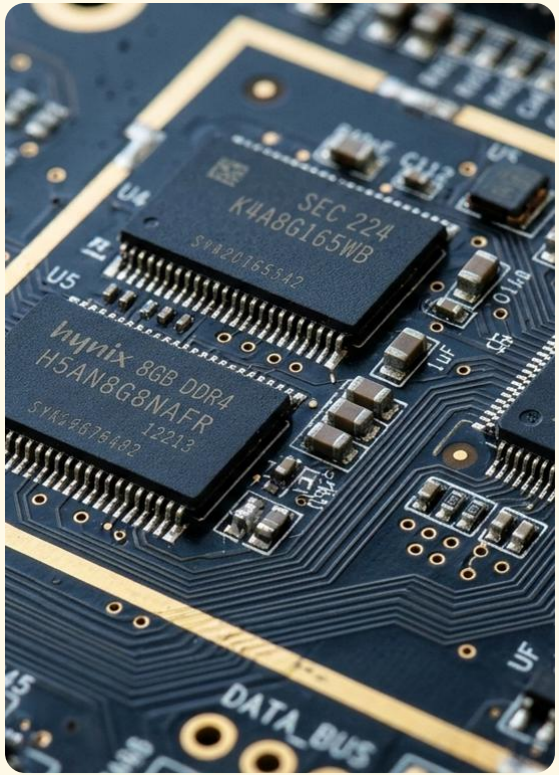
Presented by: Member 2 (Nikhil) | Random Access Memory (SRAM & DRAM)

Static RAM (SRAM)

- Cell: 4-6 transistors (flip-flop).
- Speed: Ultra-fast (< 1 - 5 ns).
- Refresh: None required (static).
- Density: Lower bit density, costly.
- Role: CPU Caches (L1, L2, L3).

Dynamic RAM (DRAM)

- Cell: 1 transistor + 1 capacitor.
- Refresh: Periodic refresh needed.
- Speed: Moderate (50 - 100 ns).
- Density: Very high, cost-effective.
- Role: Main Computer System RAM.



Read-Only Memory (ROM)

Presented by: Member 2 (Nikhil) | Non-Volatile Semiconductor Storage

ROM Characteristics

- Non-Volatile: Retains state without power.
- Bootstrapping: Houses BIOS / UEFI firmware.
- Embedded: Controls peripheral hardware.
- Read-Only: Read access during execution.

Evolution & Types

- Mask ROM: Hardwired at fab; non-alterable.
- PROM: User programmable once.
- EPROM: UV-light erased and rewritten.
- EEPROM: Electrically erased byte-by-byte.
- Flash: Block-erasable, high-density EEPROM.



Cache Memory Architecture

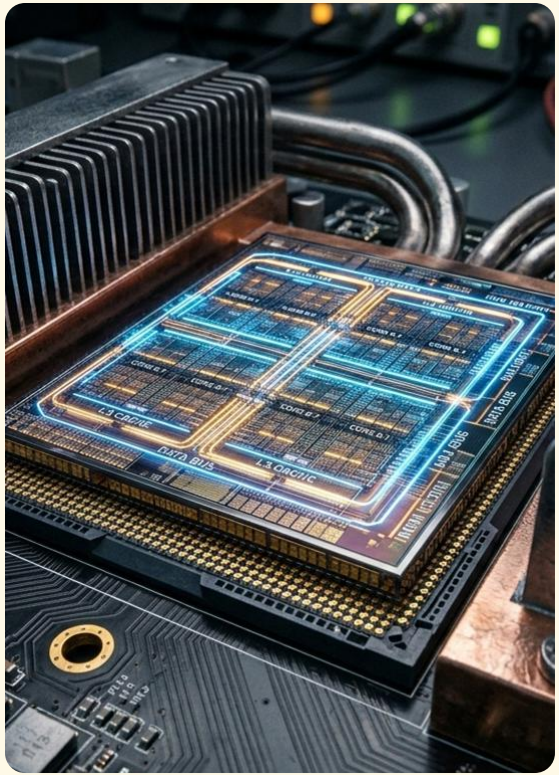
Presented by: Member 3 (Teja) | High-Speed Buffer & Multi-Level Cache

Principles of Cache

- Fast SRAM buffer between CPU & DRAM.
- **Temporal Locality:** Recently used data reused soon.
- **Spatial Locality:** Nearby addresses fetched together.
- **Transparent:** Managed automatically by hardware.

Multi-Level Hierarchy

- **L1: Dedicated per-core (< 1 ns, 32-64 KB).**
- **L2: Per-core buffer (256 KB - 1 MB).**
- **L3: Shared across all CPU cores (8 - 64 MB).**
- **Search Order: Check L1 -> L2 -> L3 -> DRAM.**



Cache Performance Considerations

Presented by: Member 3 (Teja) | Metrics & Access Time Analysis

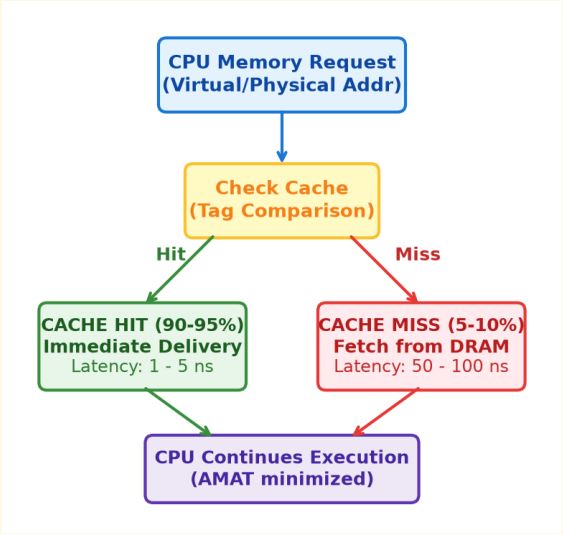
Key Performance Metrics

- **Hit:** Found in cache; Miss: Fetched from DRAM.
- **Hit Ratio (h):** Hits / Total Accesses (~90-95%).
- **Miss Penalty:** Extra latency to fetch block from RAM.
- AMAT = Hit Time + (Miss Rate × Miss Penalty).

Target Cache Hit Ratio

90% - 95%

Ensures CPU rarely experiences full DRAM latency.



Virtual Memory Architecture

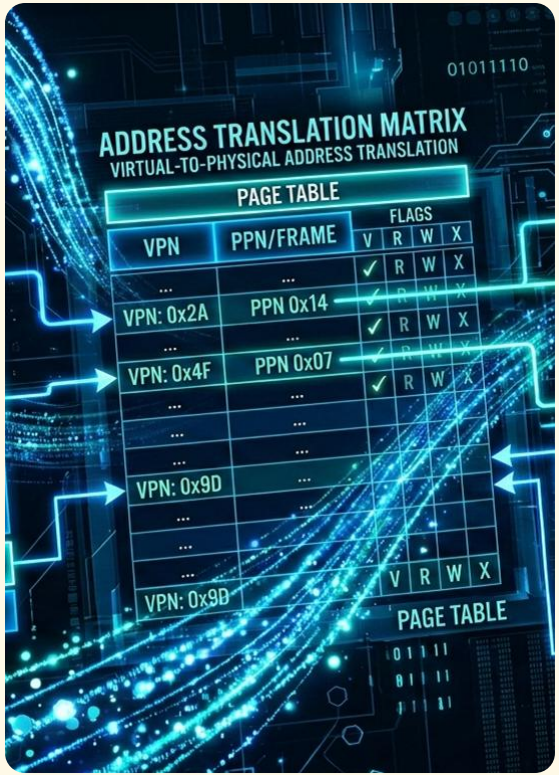
Presented by: Member 4 (Srinivas) | Address Translation & Paging

Concept & MMU

- **Logical Space: Programs see continuous memory.**
- **Beyond RAM: Runs apps larger than physical memory.**
- **MMU: Translates virtual addresses to physical RAM.**
- **Isolation: Strict per-process protection.**

Paging & Faults

- **Paging:** Virtual pages map to physical frames.
- **Page Table:** Stores virtual-to-physical translations.
- **Page Fault:** Trap when accessed page is absent.
- **Demand Paging:** OS swaps pages between RAM & disk.



Memory Management Requirements

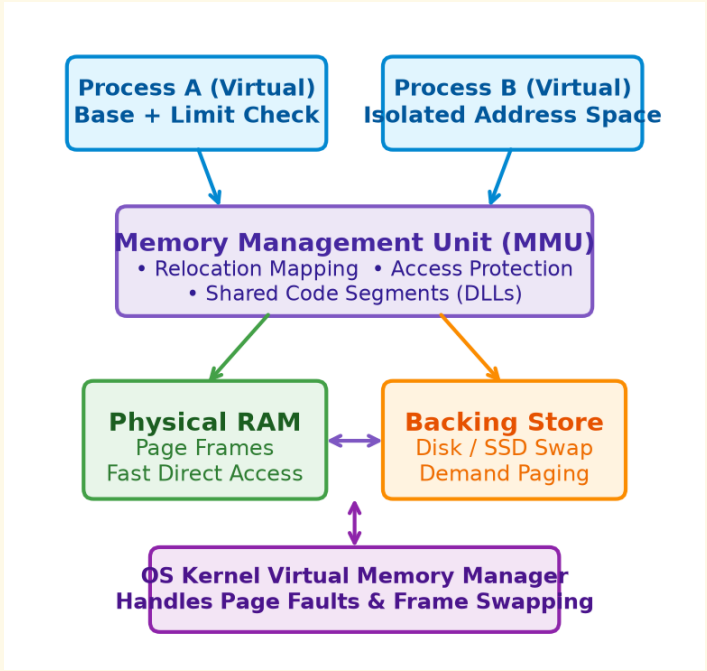
Presented by: Member 4 (Srinivas) | Core Operating System Functions

Relocation & Protection

- **Relocation:** Program executes anywhere in RAM.
- **Protection:** Prevents cross-process corruption.
- **Hardware:** Base & limit register verification.

Sharing & Modularity

- **Sharing:** Common libraries (DLLs) shared in RAM.
- **Modularity:** Organized into functions & data modules.
- **Efficiency:** Maximizes physical memory density.



Secondary Storage Systems

Presented by: Member 5 (Manohar) | Non-Volatile Mass Storage

Storage Role in Hierarchy

- **Non-Volatile:** Retains OS, binaries & data persistently.
- **Massive:** Gigabytes to terabytes at lowest cost/bit.
- **Swap Store:** Backs virtual memory page swapping.

Storage Technologies

- **SSD: NAND flash; silent, ultra-fast random I/O.**
- **HDD: Magnetic platters; high capacity & cheap.**
- **Removable: Flash USB & optical archival media.**



Comparison of Memory Levels

Presented by: Member 5 (Manohar) | Quantitative Summary Across Hierarchy

Memory Level	Typical Technology	Access Speed	Typical Capacity	Cost per Bit
CPU Registers	Flip-Flops (CMOS)	Fastest (< 1 ns)	Bytes to Kilobytes	Highest
Cache Memory	Static RAM (SRAM)	Very Fast (1-10 ns)	Kilobytes to Megabytes	Very High
Main Memory	Dynamic RAM (DRAM)	Moderate (50-100 ns)	Gigabytes (8-64 GB)	Moderate
Secondary Storage	SSD Flash / HDD Magnetic	Slower (Micro to Milliseconds)	Terabytes (512 GB - 4 TB)	Lowest

Key Takeaway on Memory Trade-offs

- **Hierarchy Rule:** Downward = speed drops ~1000x, capacity multiplies, cost/bit falls drastically.
- **Outcome:** Systems achieve near-SRAM speed with terabytes of affordable storage.

Summary & Conclusion

Presented by: Member 5 (Manohar) | Key Insights & Wrap-up

Seminar Synthesis

- Hierarchy bridges the processor-memory speed gap.
- Caches mask DRAM delay; VM enables massive address spaces.
- Delivers optimal price-performance for modern computing.

THANK YOU!



Seminar Presentation Team (CAI 2-1)

- **Team Members:** Karthikeya (M1) • Nikhil (M2) • Teja (M3) • Srinivas (M4) • Manohar (M5)
- **Subject:** Digital Logic and Computer Organization (DLCO) | Department of Computer Science & Engineering